Juliette Bruce's Future Plans

My research interests lie in algebraic geometry and commutative algebra, which are two fields that study the same objects from opposite sides. A parabola in the plane can be described geometrically as DINIS, or algebraically, as the solutions to the equation $y = x^2$. Each of these descriptions provides differing perspectives on the same questions. For example, the face that a straight line intersect a parabola in at mot two points (geometry) is precisely because a quadratic equation has at most two solutions (algebra). This dictionary between algebra and geometry is at the heart of my field, and motivates much of my research. Questions about shapes (lines, parabalas, curves, surface, etc.) becomes questions about polynomial equations and vice versa.

A running theme throughout much of my research is that the algebra of the polynomial equations defining shape capture geometry in subtle ways. For example, the equations defining a shape are rarely independent, but normally satisfy relations among themselves, called syzygies (from the Greek for "yoked together"). THIS IS GOOD BUT DO NOT MKAKE TOPRICAL STUFF A SIDE STORY

Over the next several years, my research program is to carry these tr

Syzygies of Stacky Curves. Curves are the among the most classical objects of studied in algebraic geometry and mathematics more broadly. A forty-year old program begun by Mark Green, and culminating in celebrated work of Claire Voisin, showed that the syzygies of a curve, in its most natural embedding, can be used to detect how special the curve is. The precise conjecture, known as Green's conjecture, is one of the jewels of the field.

Stacky curves are curves where extra symmetries are encoded by allowing a finite number of special points on the curve FINISH. Such spaces re not exotic and arise naturally, for example, the spaces that organize modular forms are stacky curves. Despite this prevalence, the classical theory of syzygies does not apply to them as they require variables whose degrees are not one. The equations of stacky curves were only recently described by Voight and Zureick-Borwn, and their syzygies are uncharted.

With M. Banks and P. Wei, I am building this missing theory. We have already found the overall shape of FINISH. This has led us to a stacky Green's conjecture: the special points shift the classical answer in a predictable way (exactly the number of extra equations they require). In the next several years I hope to develop the tools needed to prove this conjecture. I plan to begin by considering the simplest possible stacky curve where there is is only one spacial point.

Multigraed Castelunuov–Mumford Regularity. Classically, the Castelunuov–Mumford regularity of an algebraic variety is a single number that measures the complexity of the shape. It turns out to be central to many FIHSIN. This prevelance is underlined by a celebrated 1984 theorem of Eisenbud and Goto shows that three very measure of complexity that are on their face very different (thourgh geometyr, through syzygies, and through the degrees of the equations) all give the same answer; the Castelunuov–Mumford regularity.

My plans aim to understand and tame this behavior. First, I have conjectures that the infinite staircases are an artifact of measuring complexity with NEDED. Second, FINISH

Finally, and more speculatively, I will explore whether regularity is asking the right question, but in the wrong algebraic setting. Recent, work has shown that the algebra better captures the geometry of toric varieties when one studies a toric variety and all of its close relatives all at one. Inspired by this I am exploring whether a similar phenomena is true for regularity. ADD MORE WORDS ABOUT THE THEORY. If sucessful this theory would resolve Oda's conjecture, a long standing open probelm in toric and polyhedral geometry. SAY WHAT IT IS

Topical Geometry of Moduli Spaces. A *moduli space* is a space whose points themselves represents shapes and nearby points represent similar shapes. Despite this, many of the basic questions about the topology of such spaces remain unknown. For example, Gurshevsky asked where the moduli space of abelian varieties has nonzero odd cohomology classes – roughly the number of holes a shape has in each dimension.

Tropical geometry offers a way in; it replaces the complicated objects with their combinatorial shadows, curves are replaced with graphs, abelian varieties are replaced with polytopes, and so on. Building on breakthrough work of Chan, Galatius, and Payne for curves, my collaborators and I adapted these methods to abelian varieties. This provided numerous new cohomology classes, and positively answered Gurshevsky

question (*Geometry & Topology*, 2024). Since this moduli space is closely related to the group of symmetries of integer matrices, our relates also give new information about these groups.

A central aspect of my research program over the next several years is to develop these methods into general tools. WEIGHT TWO FOR AVS, CONE STACKS, MAP FROM SPACE TO TROPICALIZZATION REALIZING WEIGHT ZERO

Computation, Open-Source Software, & Community Center AI. Computing examples TO DEVELOP CONJECUTRES AND SURVEY THE FIELDS is a central driver underlying much of my theoretical work. For example, using high-performance high-throughoutput computing my collaborators and I computed the syzygies of hundreds of new surfaces byeond previous reaach (*Expiremental Mathematics / Journal of Algebra* ADD). Further, I have co-authored four software packages for the open-source computer algebra system *Macauly2*. I plan to coninue building upon this work in three ways. First, since no software exsits for computing with stcky curves, M. Banks, J. Cobb, and I are develop `toricStacks`, a packge for *Macualay2*. Second, I am building a searchable database of thousands of expert-chosen mathematical objects and examples (ideals, rings, modules, etc.) that are hidden in the source code and documentation of *Macauly2*. By making these examples easily accesibly this databes will make testing conjecture, benchmarking software, and developing community centered AI easier. This, with J. Voight I am planning a workshop focused on the future of mathematical software in the age of AI.